

4(a)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
	$E = \frac{1}{2} \times 29 \times (8.0 \times 10^{-2})^2$ or $E = \frac{1}{2} \times 2.32 \times 8.0 \times 10^{-2}$ $E = 9.3 \times 10^{-2} \text{ J}$	A1
4(b)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$	C1
	$= 4.5 \times 10^{-2} \times 9.81 \times 8.0 \times 10^{-2} \sin 15^\circ$	C1
	$= 9.1 \times 10^{-3} \text{ J}$	A1
4(b)(ii)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$(9.3 \times 10^{-2} - 9.1 \times 10^{-3}) = \frac{1}{2} \times 4.5 \times 10^{-2} \times v^2$	C1
	$v = (2 \times 8.4 \times 10^{-2} / 4.5 \times 10^{-2})^{0.5}$ $= 1.9 \text{ m s}^{-1}$	A1
4(c)(i)	$1.7 \times 2.0 (\times 10^{-2})$ or $4.5 \times 10^{-2} \times 9.81 \times d$	C1
	$1.7 \times 2.0 \times 10^{-2} = 4.5 \times 10^{-2} \times 9.81 \times d$ $d = 7.7 \times 10^{-2} \text{ m}$	A1
4(c)(ii)	clockwise	B1